

Assignment Problem – Classroom Exercise Sheet

This sheet contains three classroom exercises on Assignment Problems: 1. Basic Minimization 2. Maximization (converted to minimization) 3. Unbalanced Assignment Problem Students are required to solve these using the Hungarian Method.

Exercise 1: Basic Minimization

Three workers are to be assigned to three jobs. Assign each worker to one job to minimize total cost.

Worker \ Job	J1	J2	J3
W1	4	2	3
W2	2	3	1
W3	3	4	2

Exercise 2: Maximization Problem

Three workers are to be assigned to three jobs. The profit of each assignment is given below. Assign one worker per job to maximize total profit. (Hint: Convert to minimization problem first using: New Cost = Max Value - Original Value)

Worker \ Job	J1	J2	J3
W1	4	2	3
W2	2	3	1
W3	3	4	2

Exercise 3: Unbalanced Assignment Problem

Three workers are to be assigned to five jobs. Assign one worker per job to minimize total cost. (Hint: Add dummy workers with zero cost to balance the matrix)

Worker \ Job	J1	J2	J3	J4	J5
W1	9	2	7	8	6
W2	6	4	3	7	5
W3	5	8	1	8	6